

DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKENDDEVELOPMENT

DBMS_Lab_Manual_SQL_Joins_Transactions

Roll.No - 2520030429

```
Command Prompt - mysql -u x + v
Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Himavanth Sai Ganesh>cd "C:\Program Files\MySQL\MySQL Server 8.0\bin"

C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u root -p
Enter password: ****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 8.0.46 MySQL Community Server - GPL

Copyright (c) 2000, 2026, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE DBMS_LAB;
Query OK, 1 row affected (0.01 sec)

mysql> USE DBMS_LAB;
Database changed
mysql> CREATE TABLE roles(
  -> role_id INT PRIMARY KEY,
  -> role_name VARCHAR(50)
  -> );
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE users(
  -> user_id INT PRIMARY KEY,
  -> name VARCHAR(50),
  -> email VARCHAR(50),
  -> role_id INT
  -> );
Query OK, 0 rows affected (0.08 sec)

Command Prompt - mysql -u x + v

mysql> CREATE TABLE departments(
  -> dept_id INT PRIMARY KEY,
  -> dept_name VARCHAR(50)
  -> );
Query OK, 0 rows affected (0.10 sec)

mysql> CREATE TABLE employees(
  -> emp_id INT PRIMARY KEY,
  -> emp_name VARCHAR(50),
  -> dept_id INT,
  -> manager_id INT,
  -> salary DECIMAL(10,2)
  -> );
Query OK, 0 rows affected (0.09 sec)

mysql> CREATE TABLE customers(
  -> customer_id INT PRIMARY KEY,
  -> customer_name VARCHAR(50),
  -> city VARCHAR(50)
  -> );
Query OK, 0 rows affected (0.09 sec)

mysql> CREATE TABLE orders(
  -> order_id INT PRIMARY KEY,
  -> customer_id INT,
  -> order_date DATE,
  -> total_amount DECIMAL(10,2)
  -> );
Query OK, 0 rows affected (0.05 sec)

mysql> CREATE TABLE accounts(
  -> account_no INT PRIMARY KEY,
  -> holder_name VARCHAR(50),
  -> balance DECIMAL(10,2)
  -> );
Query OK, 0 rows affected (0.08 sec)
```

```

Command Prompt - mysql -u x + v
mysql> INSERT INTO roles VALUES
-> (1,'Admin'),
-> (2,'Manager'),
-> (3,'Employee'),
-> (4,'HR');
Query OK, 4 rows affected (0.02 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql> INSERT INTO users VALUES
-> (101,'Ramesh','ramesh@gmail.com',1),
-> (102,'Suresh','suresh@gmail.com',2),
-> (103,'Priya','priya@gmail.com',3),
-> (104,'Kiran','kiran@gmail.com',NULL);
Query OK, 4 rows affected (0.01 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql> INSERT INTO departments VALUES
-> (1,'CSE'),
-> (2,'ECE'),
-> (3,'MECH');
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> INSERT INTO employees VALUES
-> (1,'Anil',1,NULL,80000),
-> (2,'Sunil',1,1,60000),
-> (3,'Rahul',2,1,45000),
-> (4,'Meena',3,2,55000),
-> (5,'Divya',1,2,70000);
Query OK, 5 rows affected (0.06 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> INSERT INTO customers VALUES
-> (1,'Arun','Hyderabad'),
-> (2,'Bhanu','Vijayawada'),
-> (3,'Charan','Warangal');
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

```

```

Command Prompt - mysql -u x + v
mysql> INSERT INTO orders VALUES
-> (1001,1,'2026-06-01',15000),
-> (1002,2,'2026-06-02',20000),
-> (1003,1,'2026-06-05',10000),
-> (1004,3,'2026-06-07',25000);
Query OK, 4 rows affected (0.01 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql> INSERT INTO accounts VALUES
-> (101,'Ravi',50000),
-> (102,'Kumar',30000),
-> (103,'Sita',45000);
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> SELECT emp_name,dept_name
-> FROM employees e
-> INNER JOIN departments d
-> ON e.dept_id=d.dept_id;
+-----+-----+
| emp_name | dept_name |
+-----+-----+
| Anil     | CSE       |
| Sunil    | CSE       |
| Rahul    | ECE       |
| Meena    | MECH      |
| Divya    | CSE       |
+-----+-----+
5 rows in set (0.00 sec)

```

```
Command Prompt - mysql -u X + v

mysql> SELECT name,email,role_name
-> FROM users u
-> LEFT JOIN roles r
-> ON u.role_id=r.role_id;
+-----+-----+-----+
| name | email | role_name |
+-----+-----+-----+
| Ramesh | ramesh@gmail.com | Admin |
| Suresh | suresh@gmail.com | Manager |
| Priya | priya@gmail.com | Employee |
| Kiran | kiran@gmail.com | NULL |
+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> SELECT name,role_name
-> FROM users u
-> RIGHT JOIN roles r
-> ON u.role_id=r.role_id;
+-----+-----+
| name | role_name |
+-----+-----+
| Ramesh | Admin |
| Suresh | Manager |
| Priya | Employee |
| NULL | HR |
+-----+-----+
4 rows in set (0.00 sec)
```

```
Command Prompt - mysql -u X + v

mysql> SELECT name,email,role_name
-> FROM users u
-> LEFT JOIN roles r
-> ON u.role_id=r.role_id
-> UNION
-> SELECT name,email,role_name
-> FROM users u
-> RIGHT JOIN roles r
-> ON u.role_id=r.role_id;
+-----+-----+-----+
| name | email | role_name |
+-----+-----+-----+
| Ramesh | ramesh@gmail.com | Admin |
| Suresh | suresh@gmail.com | Manager |
| Priya | priya@gmail.com | Employee |
| Kiran | kiran@gmail.com | NULL |
| NULL | NULL | HR |
+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> SELECT e.emp_name AS Employee,
-> m.emp_name AS Manager
-> FROM employees e
-> LEFT JOIN employees m
-> ON e.manager_id=m.emp_id;
+-----+-----+
| Employee | Manager |
+-----+-----+
| Anil | NULL |
| Sunil | Anil |
| Rahul | Anil |
| Meena | Sunil |
| Divya | Sunil |
+-----+-----+
5 rows in set (0.00 sec)
```

```

Command Prompt - mysql -u × + v

mysql> SELECT o.order_id,
->         customer_name,
->         city,
->         total_amount
-> FROM customers c
-> JOIN orders o
-> ON c.customer_id=o.customer_id;
+-----+-----+-----+-----+
| order_id | customer_name | city       | total_amount |
+-----+-----+-----+-----+
| 1001     | Arun          | Hyderabad | 15000.00     |
| 1002     | Bhanu         | Vijayawada | 20000.00     |
| 1003     | Arun          | Hyderabad | 10000.00     |
| 1004     | Charan        | Warangal   | 25000.00     |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> SELECT dept_name,
->         COUNT(*) AS EmployeeCount,
->         AVG(salary) AS AverageSalary,
->         MAX(salary) AS MaximumSalary
-> FROM employees e
-> JOIN departments d
-> ON e.dept_id=d.dept_id
-> GROUP BY dept_name;
+-----+-----+-----+-----+
| dept_name | EmployeeCount | AverageSalary | MaximumSalary |
+-----+-----+-----+-----+
| CSE       | 3             | 70000.000000  | 80000.00      |
| ECE       | 1             | 45000.000000  | 45000.00      |
| MECH      | 1             | 55000.000000  | 55000.00      |
+-----+-----+-----+-----+
3 rows in set (0.05 sec)

```

```

Command Prompt - mysql -u × + v

mysql> SELECT emp_name,salary,dept_name
-> FROM employees e
-> JOIN departments d
-> ON e.dept_id=d.dept_id
-> WHERE salary>50000
-> AND dept_name='CSE';
+-----+-----+-----+
| emp_name | salary  | dept_name |
+-----+-----+-----+
| Anil     | 80000.00 | CSE       |
| Sunil    | 60000.00 | CSE       |
| Divya    | 70000.00 | CSE       |
+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> SELECT customer_name,
->         SUM(total_amount) AS TotalPurchase
-> FROM customers c
-> JOIN orders o
-> ON c.customer_id=o.customer_id
-> GROUP BY customer_name
-> ORDER BY TotalPurchase DESC;
+-----+-----+
| customer_name | TotalPurchase |
+-----+-----+
| Arun          | 25000.00     |
| Charan        | 25000.00     |
| Bhanu         | 20000.00     |
+-----+-----+
3 rows in set (0.00 sec)

mysql> CREATE TABLE faculty(
->     faculty_id INT PRIMARY KEY,
->     faculty_name VARCHAR(50)
-> );
Query OK, 0 rows affected (0.04 sec)

```

```

mysql>
mysql> CREATE TABLE courses(
->     course_id INT PRIMARY KEY,
->     course_name VARCHAR(50),
->     faculty_id INT
-> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> CREATE TABLE students(
->     student_id INT PRIMARY KEY,
->     student_name VARCHAR(50)
-> );
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> CREATE TABLE enrollments(
->     enrollment_id INT PRIMARY KEY,
->     student_id INT,
->     course_id INT
-> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO faculty VALUES
-> (1,'Dr Rao'),
-> (2,'Dr Kumar');
Query OK, 2 rows affected (0.05 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO courses VALUES
-> (101,'DBMS',1),
-> (102,'Java',2);
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0

```

```

mysql>
mysql> INSERT INTO students VALUES
-> (1,'Ganesh'),
-> (2,'Sai');
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO enrollments VALUES
-> (1,1,101),
-> (2,2,102);
Query OK, 2 rows affected (0.07 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> SELECT student_name,
->     course_name,
->     faculty_name
-> FROM students s
-> JOIN enrollments e
-> ON s.student_id=e.student_id
-> JOIN courses c
-> ON e.course_id=c.course_id
-> JOIN faculty f
-> ON c.faculty_id=f.faculty_id;
+-----+-----+-----+
| student_name | course_name | faculty_name |
+-----+-----+-----+
| Ganesh      | DBMS       | Dr Rao      |
| Sai         | Java       | Dr Kumar    |
+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> INSERT INTO accounts
-> VALUES(104,'Anil',25000);
Query OK, 1 row affected (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

```

Command Prompt - mysql -u × + ▾

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> INSERT INTO accounts
-> VALUES(105,'Ravi',10000);
Query OK, 1 row affected (0.00 sec)

mysql> ROLLBACK;
Query OK, 0 rows affected (0.05 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE accounts
-> SET balance=balance-5000
-> WHERE account_no=101;
Query OK, 1 row affected (0.05 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql>
mysql> UPDATE accounts
-> SET balance=balance+5000
-> WHERE account_no=102;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE accounts
-> SET balance=balance-3000
-> WHERE account_no=101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

Command Prompt - mysql -u × + ▾

```
mysql>
mysql> UPDATE accounts
-> SET balance=balance+3000
-> WHERE account_no=999;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0 Changed: 0 Warnings: 0

mysql>
mysql> ROLLBACK;
Query OK, 0 rows affected (0.05 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> UPDATE accounts
-> SET balance=balance+2000
-> WHERE account_no=101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql>
mysql> SAVEPOINT s1;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE accounts
-> SET balance=balance-1000
-> WHERE account_no=101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql>
mysql> ROLLBACK TO s1;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.05 sec)
```